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2026 IEEE 5th World Conference on Applied Intelligence and Computing : Submission (3088) has been edited.

1 message

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Mon, May 4, 2026 at 11:39 PM

Hello,

The following submission has been edited.

Track Name: AIC2026

Paper ID: 3088

Paper Title: A Comparative Study of Infrastructure Automation Tools in Cloud Environments

Abstract:

Cloud computing has transformed how organizations design, deploy, and manage IT infrastructure, making Infrastructure as Code (IaC) and automation essential components of modern DevOps practices. This study presents a systematic comparative evaluation of three widely used infrastructure automation tools: Ansible, Terraform, and Puppet. A controlled, simulation-based experimental methodology was used to execute identical infrastructure tasks across varying node scales (5, 10, 25, and 50) in an AWS environment.

The results show that Ansible consistently outperforms the other tools in deployment speed and resource efficiency, completing a 10-node deployment 38% faster than Terraform and 54% faster than Puppet, while also consuming the least CPU and memory. Terraform demonstrates strong capabilities in stateful infrastructure provisioning and multi-cloud support, whereas Puppet excels in configuration compliance and drift detection for enterprise environments.

A weighted qualitative evaluation further ranks Ansible as the highest-performing tool overall (45/50), followed by Terraform (38/50) and Puppet (26/50). The findings provide practical, evidence-based recommendations for selecting appropriate automation tools in cloud-native DevOps workflows.

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Submission Questions Response:

1. Conflict of Interest
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all; supervision: Isha Kansal

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